

JYOTHIR RAGHAVALU BHOGI

Generative AI Engineer · LLMs, RAG & Agentic AI Systems

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SUMMARY

Generative AI Engineer with two AI/ML internships and three independently built, production-oriented GenAI systems — RAG pipelines, multi-agent orchestration, and LLM evaluation workflows, each exposed through REST APIs rather than left as notebooks. Reduced manual data-verification effort by 70% and improved ML pipeline execution speed by 25% during a data engineering internship. B.Tech Computer Science candidate (2026) focused on engineering reliability — retrieval, orchestration, and evaluation — around large language models.

CORE COMPETENCIES

Generative AI: LLMs, RAG, Prompt Engineering, AI Agents (LangChain, LangGraph), Embeddings & Vector Search (Qdrant)
ML & Engineering: Python, FastAPI, PyTorch, Hugging Face, ML Pipelines, Model Evaluation, AWS, Docker, PySpark, SQL

EXPERIENCE

- Generative AI Engineer Intern — Know AI**

Jun 2026 – Aug 2026

 - Designed and built RAG pipelines using embeddings and vector search, improving contextual retrieval and output reliability for LLM applications.
 - Built backend REST APIs (FastAPI) integrating LLM inference, search, and workflow automation into unified services.
 - Developed AI/ML evaluation pipelines for model testing, error analysis, and output validation, catching regressions before deployment.
 - Applied prompt engineering and structured evaluation to Generative AI workflows, improving output reliability and consistency.
- Data Engineer Intern — flickDONE**

Jun 2025 – Aug 2025

 - Built Python-based ML data pipelines processing 50,000+ records for preprocessing, training, and validation.
 - Reduced manual verification effort by 70% by engineering automated preprocessing and data-validation workflows.
 - Improved AWS ML pipeline execution speed by 25% through targeted bottleneck analysis.
 - Developed reusable AI workflow components for conversational AI, document processing, and business automation.

SELECTED PROJECTS

- Production-Grade RAG Engine** — Python, FastAPI, LangChain, Qdrant, AWS

Jun 2024 – Aug 2024

 - Built an end-to-end RAG pipeline: document ingestion, chunking, embedding, vector retrieval, and context generation.
 - Implemented semantic search via a Qdrant vector database for efficient context retrieval.
 - Added reranking, retrieval evaluation, fallback execution, and monitoring to improve reliability and response quality.
- LangGraph Workflow Orchestrator** — Python, LangGraph, LangChain, FastAPI, Gemini

Aug 2025

 - Built a stateful multi-agent orchestration platform with state management, conditional routing, and modular task decomposition.
 - Exposed workflow execution through FastAPI REST APIs for reusable AI task automation.
 - Improved reliability through checkpointing, failure handling, testing, and execution monitoring.
- Regulatory Compliance RAG Assistant** — Python, Streamlit, LangChain, Embeddings

Apr 2026 – Jun 2026

 - Built a Generative AI compliance assistant with document ingestion, semantic retrieval, and context-grounded generation.
 - Developed interactive document upload and querying workflows using Streamlit.
 - Improved response reliability through retrieval testing, evaluation, and debugging.

EDUCATION

Lovely Professional University
Bachelor of Technology, Computer Science & Engineering

Jalandhar, India
Jul 2022 – May 2026

CERTIFICATIONS

Generative AI with Large Language Models — Coursera

2024

Python, Data Science & Machine Learning — CipherSchools

2024